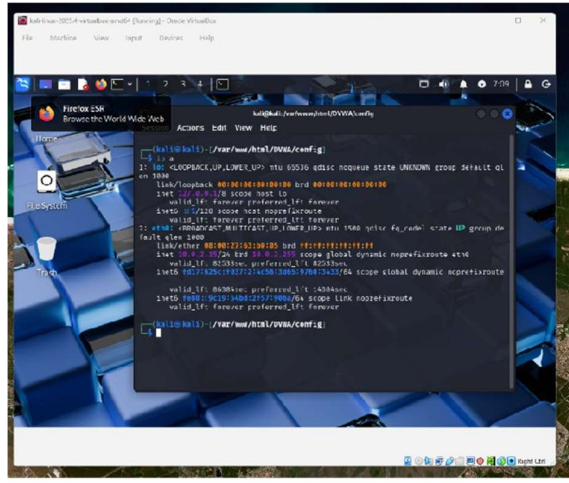


Practical 02

Important: Network scanning should be done only on authorized systems. Do not scan public or unknown networks.

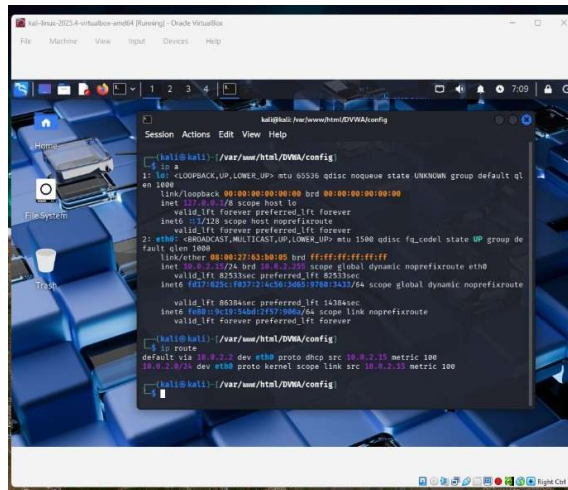
1. Open Terminal. Run the following command
 - a. *ip a* : Used to display IP address and network interface details.



```
(kali@kali):~/var/www/html/DVWA/config$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN scope default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 10.0.2.15/24 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:b4:00 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 scope global dynamic noprefixroute eth0
        valid_lft 85536sec preferred_lft 85536sec
    inet6 fd17:025c:f037:24c5b:3080:9708:3a32:64 scope global dynamic noprefixroute
        valid_lft 86384sec preferred_lft 14384sec
    inet6 fe80::9c15:304d:2397:990a:6a scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Expected: IP address (e.g., 10.0.2.15) should be displayed.

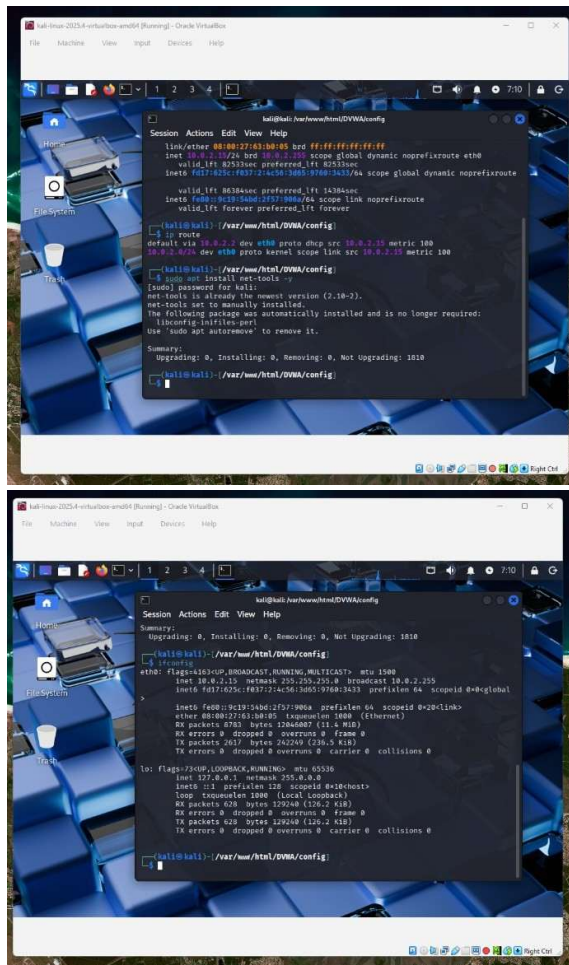
2. Check Default Gateway.
 - a. Run the following command:
ip route



```
(kali@kali):~/var/www/html/DVWA/config$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
```

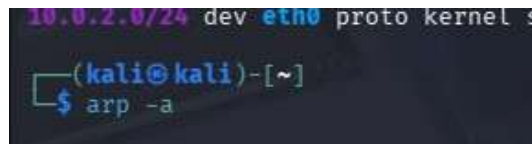
3. Running ifconfig: Run the following commands:
 - a. *sudo apt install net-tools -y*
 - b. *ifconfig*

Note: ifconfig is a legacy command used to view network configuration. It is not installed by default in modern systems.



4. View ARP Table

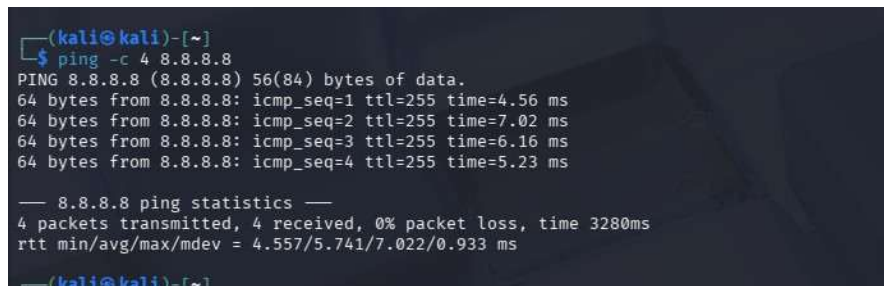
- Run `arp -a`



Note: If no entries are shown, first run the ping command and then execute `arp -a` again.

5. Test Connectivity.

- Run `ping -c 4 8.8.8.8`

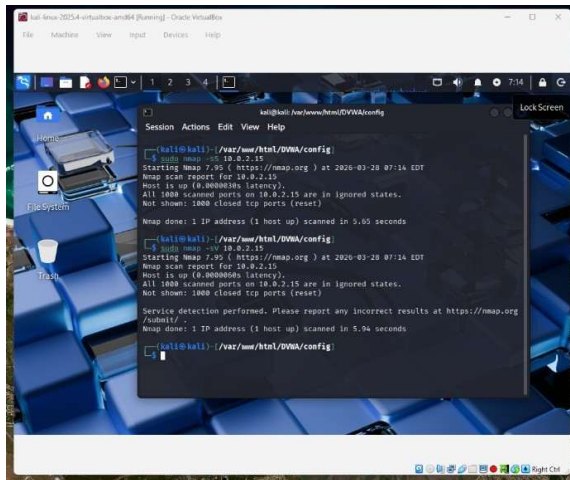


Expected: Replies should be received with 0% packet loss.

6. Scan Open Ports of Target System. Run the following commands:

- `sudo nmap -sS <target-ip>`
- `sudo nmap -sV <target-ip>`

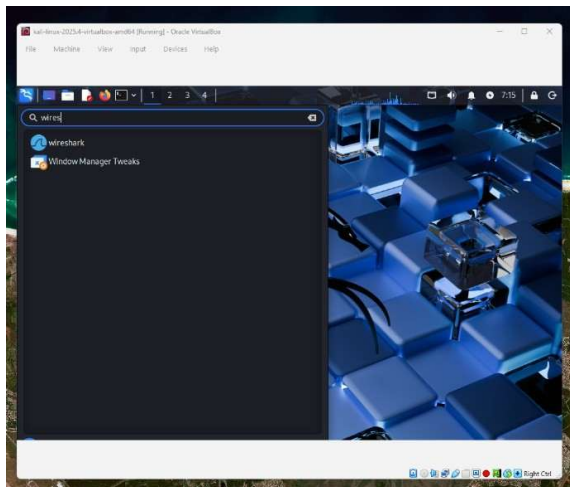
Replace <target-ip> with the IP address obtained using ip a.
Performs a TCP SYN scan to identify open ports.



Expected: List of open ports and services will be displayed.

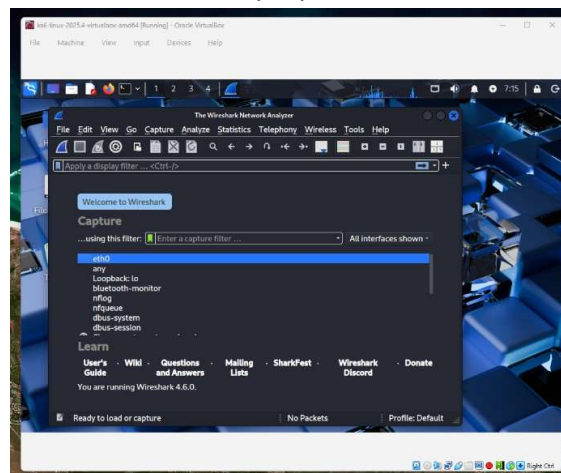
7. Packet Analysis

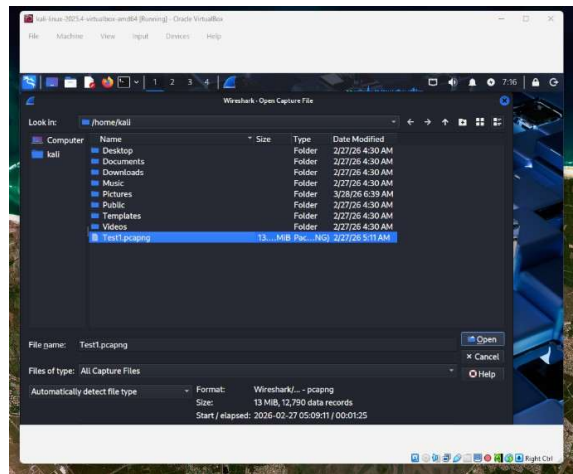
a. Open Wireshark



b. Open pcap File

- Click File → Open
- Select downloaded .pcap file



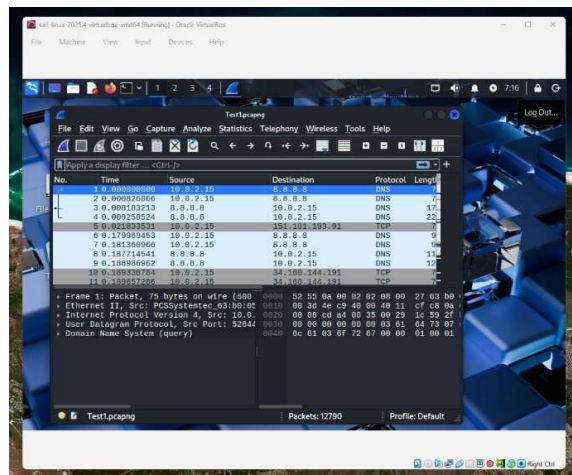


Note: Use a sample .pcap file downloaded from a trusted source

c. Analyze Traffic

i. Look for:

1. TCP handshake (SYN, SYN-ACK, ACK)
2. DNS requests
3. HTTP packets



d. Use filter examples:

- i. tcp
- ii. dns
- iii. http

